

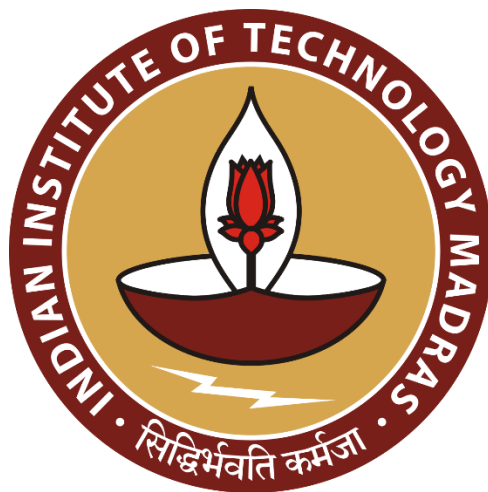
Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 1 Report

Group 9

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1. Abstract

The rise of synthetic media, such as Deep Fakes and AI-generated text, presents major risks to information integrity and digital trust. Malicious actors exploit these tools to spread disinformation, manipulate public perception, and compromise individuals' reputations. This project proposes a two-part system to address this challenge.

A vision-based classifier detects whether an uploaded image is real or manipulated using fine-tuned multimodal models such as CLIP. A text credibility pipeline processes user-submitted text by summarizing it with a large language model (LLM), retrieves evidences via web search APIs, and delivers a final credibility report.

2. Objectives

- **Deep Fake Detection** – Develop an image-based classifier to identify manipulated or synthetic images, using labeled datasets of real and fake images.
- **Text Credibility Analysis** – Enable users to submit text and receive a detailed credibility assessment through Natural Language Processing.
- **End-to-End User Experience** – Provide a simple interface for users to either submit their inputs and receive clear, actionable results.

This work primarily benefits journalists combating misinformation, educators teaching media literacy, fact-checkers, and the general public who want quick verification tools.

3. Literature Review

3.1. Existing Approaches in Image Deep Fake Detection

- **Traditional CNN-based methods** (e.g., XceptionNet, MesoNet) have been widely used for detecting fake images and videos. They achieve high accuracy on curated datasets but often fail to generalize to unseen manipulations.

- **Multimodal models like CLIP** (Contrastive Language–Image Pretraining) offer promising avenues for fine-tuning because they generalize across tasks and modalities. Fine-tuned CLIP can outperform specialized CNNs when limited labeled data is available.

Drawbacks: Many models are computationally heavy, sensitive to domain shifts, and often lack explainability for non-technical users.

3.2 Existing Approaches in Text Credibility and Fact-Checking

- **LLM-based summarization** provides a way to condense long or complex claims into verifiable statements. Models such as GPT-4 or Gemini have shown strong performance in contextual summarization.
- **Automated fact-checking pipelines** (e.g., ClaimBuster, Full Fact AI) combine claim detection, evidence retrieval, and stance detection. However, these are often brittle, domain-specific, or lack transparency for end users.

Drawbacks: Fact-checking systems often face challenges in retrieval quality, bias in LLM outputs, and scalability.

3.3 Contribution of Proposed Solution

Our system aims to integrate these insights into a dual pipeline that:

- Leverages fine-tuned CLIP for efficient, generalizable fake image detection.
- Uses LLMs + web search APIs for transparent credibility analysis of text.
- Provides a single, user-friendly interface for multimodal verification.

4. Model Architecture

4.1 Deep Fake Detection (Image-Only Submission)

- **Input:** User-submitted image.
- **Model:** Pre trained Vision Model (e.g., CLIP's Vision Transformer, EfficientNet etc.)

- **Finetuning:** The model will be finetuned on a carefully curated dataset of "real" and "fake" images, with a focus on diverse deepfake generation methods.
- **Output Layer:** A classification head (e.g., a fully connected layer with a sigmoid activation) will output a probability score indicating whether the image is a fake or not (e.g., 0 for real, 1 for fake).
- **Output:** Binary classification (Fake/Real).

4.2 Disinformation Detection (Text-Only Submission)

- **Input:** User-submitted text.
- **Component 1: LLM Summarization:**
 - **Function:** Extracts key claims, entities, and the core message from the input text.
 - **Prompt Engineering:** Careful prompt design will guide LLM to produce concise and accurate summaries suitable for web queries.
- **Component 2: Web Search with APIs:**
 - **Function:** Takes the summarized claims and performs targeted web searches using APIs (e.g., Tavily, Perplexica).
 - **Output:** A collection of search results (links, snippets, page content) from various sources.
- **Component 3: LLM Pro Credibility Report Generation:**
 - **Input:** Original text, summarized claims, and retrieved web search results.
 - **Function:** Gemini analyzes the provided information to:
 - Compare claims against external evidence.
 - Identify factual inconsistencies or contradictions.
 - Assess source credibility based on search results.
 - Highlight emotionally charged or biased language.
 - Provide contextual information.
 - Generate a structured report outlining the credibility of the text.

- **Output:** A detailed credibility report, potentially including a numerical score or qualitative assessment (e.g., "Highly Credible," "Partially True," "Misleading," "False").

5. Dataset

To support the image forgery detection pipeline, we will rely on publicly available labeled image datasets that provide clear ground truth (real vs fake).

- **Kaggle:** [Deep Fake and Real Images Dataset](#)

This dataset contains a large set of labeled fake and real images, generated using GAN-based methods and paired with authentic samples. It is directly usable for binary classification (real vs fake), making it suitable for fine-tuning CLIP or other transformer-based vision models.

- **OpenForensics:** [Multi-Face Forgery Detection and Segmentation In-The-Wild](#)

Introduced by Le et al. (ICCV 2021), OpenForensics is a challenging large-scale dataset designed for multi-face forgery detection and segmentation in unconstrained environments. It includes manipulated and real faces in-the-wild, providing both image-level binary labels and pixel-level annotations. This dataset is particularly valuable because it captures realistic, social media-like conditions where multiple faces appear in a single image.

- **Web Evidence (for Text Credibility Pipeline)**

For text credibility assessment, we will not rely on a static dataset but instead use real-time search APIs (Tavily, Perplexica, Google Custom Search, Bing Web Search etc.). These APIs will retrieve external evidence, which the LLM will use to support or refute the input claims.

6. Feasibility and Compute

- **Compute Resources:** GPU-backed cloud instances (e.g., Google Colab Pro or AWS EC2). CLIP fine-tuning requires moderate resources (1–2 GPUs). LLM access will be via API (Gemini or OpenAI).
- **Implementation Feasibility:** Leveraging existing pretrained models and datasets reduces training time. The main challenge is integration and evaluation.
- **Ethical Considerations:** The tool will include disclaimers to prevent misuse, avoid overconfidence in model predictions, and ensure transparency.

7. Evaluation Metrics

- **Image Detection:** Accuracy, F1-score, ROC-AUC on benchmark datasets.
- **Text Credibility:** Alignment with fact-checking sources, human evaluation of reports, consistency across queries.
- **User Experience:** Usability feedback and response time.

8. Roles

- **Project Manager** – Coordination, documentation, milestone tracking.
- **Vision Lead** – Dataset preparation, fine-tuning CLIP, image evaluation pipeline.
- **NLP Lead** – LLM summarization, credibility report design.
- **Backend/Integration Lead** – Connecting pipelines with APIs, system architecture.
- **Frontend/UI Lead** – Interface design and user experience.

9. Timeline

- **Week 1-2:** Literature review, dataset curation, initial UI sketches.
- **Week 3-4:** Image pipeline setup, preliminary CLIP fine-tuning.
- **Week 5-6:** Text pipeline setup with Gemini API + web search integration.
- **Week 7-8:** Integration testing + Evaluation + Final report.

10. Conclusion

This milestone establishes the foundation for a unified system tackling misinformation in both visual and textual domains. By fine-tuning multimodal models for DeepFake detection and coupling LLM summarization with evidence retrieval for text credibility, the system addresses a pressing societal need. The team will continue to refine datasets, pipelines, and user interfaces in subsequent milestones.